



A Whole-Building Demonstration of Differentiable Predictive Control with Stochastic Occupant Behavior

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Abstract

This paper presents whole-building simulation results that demonstrate the use of differentiable predictive control (DPC) for supply-air-temperature reset in a reference office building with stochastic occupancy. A physics-informed DPC was trained on a single HVAC zone to optimize occupant comfort and HVAC energy use. When deployed on 5 zones on the same floor and compared with rule-based control strategies, DPC reduced occupied comfort violations by 54.8% and reduced total HVAC energy use by 21.3%. When the same DPC was then zero-shot deployed across all 15 zones in the entire building, it delivered a 39% reduction in comfort violations and a 14% reduction in total HVAC energy use. These results demonstrate that DPC is a scalable, occupant-centric framework that enables predictive, energy-efficient building control under real-world stochastic occupancy conditions.

Introduction

Human occupant behavior has a significant impact on building system operation and energy use, yet traditional design practices often assume deterministic occupancy (O'Brien, W. et al., 2020). In the real world, occupant behavior is inherently stochastic; neglecting this variability can lead to HVAC system over-sizing and operational inefficiencies whenever actual occupancy diverges from design assumptions. Commercial and open-source simulation tools that can generate stochastic occupancy behavior have long been available, but such tools are rarely used in building performance simulation workflows. Further, their compatibility with advanced control strategies—particularly Model Predictive Control (MPC), which relies on accurate predictions of future control inputs—remains underexplored (Elehwany et al., 2024).

Recent advances in physics-informed machine learning (PIML) open new opportunities for modeling and controlling the complex dynamics of buildings, especially when vast amounts of training data are not available. Machine learning models that rely exclusively

on training data typically demand large, high-quality datasets, may lack physical consistency, and generalize poorly to unobserved conditions. By combining physics-based models with data-driven learning, PIML can simultaneously improve interpretability, data efficiency, and robustness (Jiang et al., 2025). Moreover, differentiable predictive control (DPC)—which embeds the learned model directly within the control optimization—has shown particular promise as an end-to-end framework for building control (Drgoňa et al., 2022). Still, empirical assessments of DPC in buildings remain limited; prior studies report results only for simplified multi-zone environments (Drgoňa et al., 2021).

Accordingly, this paper (i) extends prior DPC evaluations from the building zone level to the whole-building scale, (ii) integrates stochastic occupancy simulation into PIML training, and (iii) investigates the relationship between training fidelity and energy performance. We examine two scenarios: (1) whether a PIML-based zone-level predictor trained with stochastic occupancy maintains performance across other zones with different occupant behavior, and (2) whether a DPC policy trained on one floor preserves control performance on other floors. Our evaluation leverages previous work, including (a) a high-fidelity prototype medium-sized office Building Information Model (BIM) and corresponding EnergyPlus energy model, (b) a DPC controller trained to regulate HVAC temperature setpoints, and (c) a simulation workflow that supports stochastic occupant behavior (Figure 1).

In multi-zone commercial VAV systems, supply-air temperature (SAT) reset is a control variable that balances energy efficiency and thermal comfort by coordinating trade-offs among the supply fan, cooling coil, and reheat coils. Accordingly, this study focuses on SAT control. Strategies commonly used in practice include: (1) fixed SAT setpoint (no reset), (2) outdoor-air-temperature (OAT)-based SAT reset, (3) the Trim & Respond (T&R) sequence specified in ASHRAE Guideline 36 (G36). While these approaches are

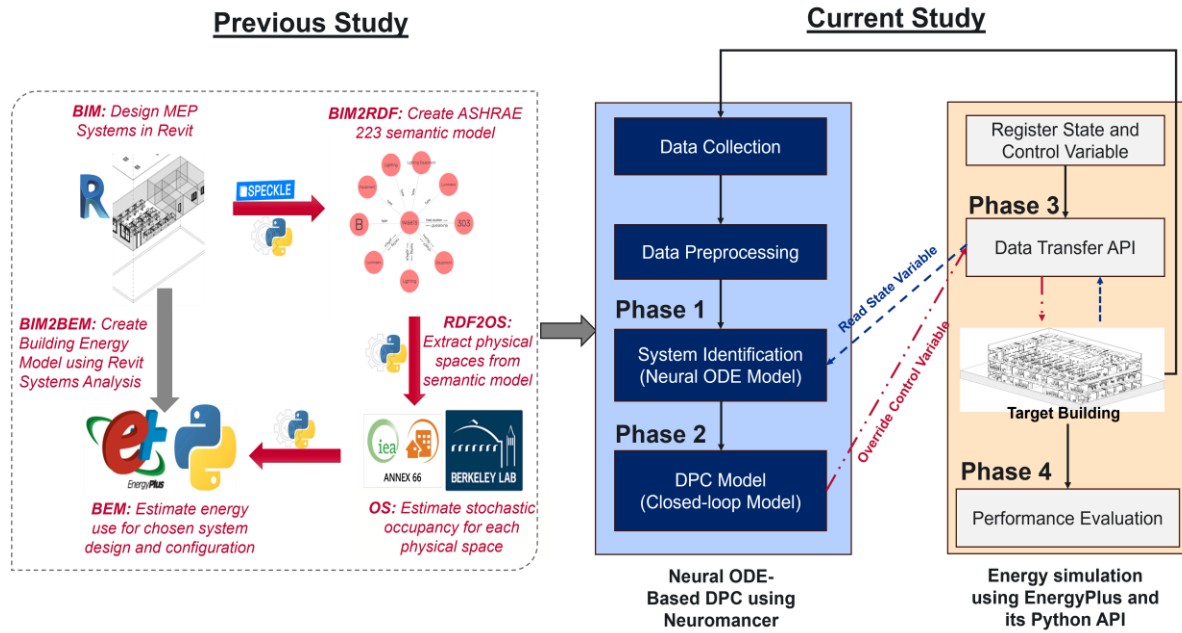


Figure 1. Research Workflow

relatively easy to implement, their robustness under uncertain operating conditions—such as load variation and occupancy fluctuation—remains insufficiently validated (Elehwany et al., 2024).

Literature Review

Recent work has actively explored predictive control techniques—MPC and Reinforcement Learning (RL)—for SAT reset. Liang et al. (2015) identified an autoregressive moving-average exogenous (ARMAX) model from field data for an AHU serving a multi-zone VAV building. They designed an MPC that respected downstream VAV load constraints. Compared to conventional PID control, their approach achieved average HVAC energy savings exceeding 27.8%, demonstrating the practical efficacy of MPC.

Building on this, Lee et al. (2019) proposed a real-time SAT optimization algorithm using an artificial neural network (ANN) in an EnergyPlus–MATLAB–building controls virtual test bed (BCVTB) co-simulation environment. The ANN predicted next-step energy consumption from inputs such as outdoor conditions, solar irradiance, and cooling load, and dynamically adjusted SAT; the method yielded up to 18.2% cooling-energy savings relative to fixed SAT control, reducing chiller energy while tempering fan-energy increases to improve overall efficiency.

RL-based approaches lessen reliance on explicit models and can learn stable policies in nonlinear, continuous control settings. Yoon et al. (2024) reported—via field and simulation studies—that RL-based SAT control

achieved approximately 10–20% HVAC energy savings over G36. In an EnergyPlus–Python co-simulation, Fang et al. (2022) employed a deep Q network (DQN) to reset SAT and chilled-water supply temperature in real time, outperforming fixed-setpoint baselines in both energy use and comfort maintenance. Using measured Building Automation System (BAS) data, Zou et al. (2020) developed a long short-term memory (LSTM)-based HVAC simulator coupled with an RL framework; the resulting controller reduced annual energy use by 27–30% while maintaining comfort.

Occupant behavior is inherently stochastic and varies across time and space in commercial buildings. Stochastic occupancy modeling frameworks can capture realistic human behavior, enabling more credible performance assessments of occupancy-based control strategies (Chen et al., 2018). Yet these tools are seldom integrated into control modeling and simulation workflows, even when control performance is known or expected to be highly sensitive to occupancy variability and forecast errors (Yang et al., 2021).

The influence of occupant behavior and zone-specific thermal preferences on SAT control performance has been investigated to a limited degree, thus far. Elehwany et al. (2024) used measured binary (0/1) occupancy time-series to derive nine zone-level occupancy profiles and ten composite scenarios, comparing four SAT reset strategies: fixed 13 °C, OAT-based reset, T&R, and hybrid T&R. They found that occupancy variability and heterogeneous temperature preferences significantly affect energy use and comfort metrics. T&R-based

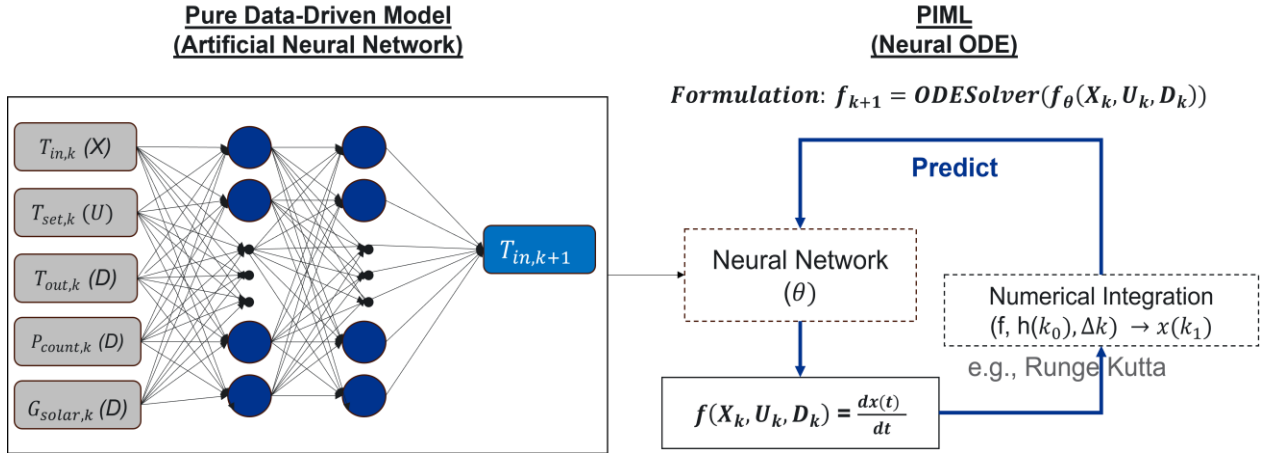


Figure 2. Neural ODE Architecture for Zone temperature predictor

strategies delivered relatively stable comfort but exhibited persistent oscillations in the SAT signal due to their reactive nature. To address this, the authors advocated optimal control formulations—such as MPC or RL—for improved SAT reset, although their evaluation was conducted on a simplified building model. In summary, while prior SAT-reset studies evaluated diverse control strategies, they used deterministic occupancy schedules, thereby underestimating the impact of stochastic occupancy on control performance. Post-COVID-19, many office buildings operate under hybrid work arrangements, resulting in more dynamic, less predictable occupancy patterns.

The energy cost-savings potential of occupancy-based control in such environments is significant, with reported savings of up to 50% (Mantesi et al., 2022). Occupant-centric control seeks to optimize energy use while maintaining thermal comfort and indoor air quality.

Motivated by these gaps, we present a predictive SAT-reset framework that explicitly accounts for stochastic occupancy patterns, evaluating not only energy savings but also occupant comfort and control robustness.

Methodology

Figure 1 summarizes how a previously developed workflow and energy use simulation is leveraged towards this work. An EnergyPlus building energy model (BEM) and an ASHRAE 223 semantic model are created from a Revit-based BIM. From this semantic model, physical spaces are extracted, and HVAC zone-level stochastic occupancy schedules are generated using Annex 66-type methods (Yan et al., 2017). These schedules are embedded in the EnergyPlus BEM, and the BEM is simulated for 12 months. The DPC predictor is trained on exactly one of these HVAC zones, and then transferred to the other zones to evaluate its performance when subject to different occupant behavior. Notably,

occupant behavior varies as a result of both the stochasticity introduced by the simulator and also the varying underlying space types in each HVAC zone.

The previously simulated energy use data is preprocessed, and in Phase 1 the building dynamics are modeled via a Neural ODE selected from the Neuromancer library (Dr̄goňa et al., 2023). In Phase 2, a DPC policy is trained and then deployed in Phase 3 within an EnergyPlus–Python API co-simulation. Finally, Phase 4 quantifies performance using energy efficiency and comfort violation metrics.

Supply Air Temperature (SAT) Reset Strategies

Traditional approaches to SAT reset rely on simplified assumptions about the SAT–total energy cost relationship, demand repeated building-specific parameter tuning, and lack continuous optimization and feedback. To mitigate these limitations, ASHRAE Guideline 36 (2018) specifies a standardized T&R sequence that considers both zone cooling demand and OAT, improving interoperability across building automation systems, yet still depends on designer-specified SAT and OAT limits. To overcome these constraints, this study applies a PIML-based DPC framework that explicitly accounts for probabilistic occupancy and dynamic load conditions, aiming to optimize SAT while simultaneously improving energy efficiency and thermal comfort.

Differentiable Predictive Control (DPC) Framework

System Identification

To enhance predictive performance under occupancy uncertainty, we adopt a Neural Ordinary Differential Equation (Neural ODE) model within a PIML framework (Chen et al., 2018). Although purely data-driven ANN can learn complex nonlinear input–output

mappings when ample data are available, they often underrepresent a building's continuous-time dynamics and instead fit statistical patterns in the data. This limitation can degrade generalization under disturbances—such as abrupt changes in outdoor air or stochastic occupancy fluctuations. By preserving physical temporal continuity, Neural ODEs mitigate these issues.

Figure 2 illustrates the Neural ODE architecture for zone-level prediction, which couples an ANN with an ODE-based state update. The ANN (Figure 2, left) is a standard multilayer-perceptron model that maps inputs to outputs in discrete time: given the current state x_k , control u_k , and disturbance d_k , it predicts the next step indoor temperature x_{k+1} . This relationship is expressed as equation (1).

$$x_{k+1} = f_{\theta, \text{ANN}}(x_k, u_k, d_k) \quad (1)$$

The composition of the input and output data used for training and evaluation is summarized in Table 1. The ODE-based state update (Figure 2, right) reformulates prediction as a continuous-time dynamical system. Rather than directly forecasting the following output, the Neural ODE learns the state derivative and integrates it forward with a numerical ODE solver to obtain future states. The continuous-time dynamics are given by equation (2):

$$\frac{dx(t)}{dt} = \text{ODESolver}(f_{\theta, \text{NODE}}(x(t), u(t), d(t))) \quad (2)$$

Here, $f_{\theta, \text{NODE}}$ is a neural network that approximates the state-rate function; future states are computed via the ODE integrator.

Table 1. Input-Output datasets

Categories	Input data	Output data
State Dataset (x_k)	Indoor Temp. ($T_{in,k}$)	
Control Dataset (u_k)	Supply Air Set Temp. ($T_{set,k}$)	N-step
	Outdoor Air Temp. ($T_{out,k}$)	Indoor Temp. ($T_{in,k+n}$)
Disturbance Dataset (d_k)	People Count Number ($P_{count,k}$)	
	Solar Irradiance ($G_{solar,k}$)	

In this work, we use the classical fourth-order Runge–Kutta (RK4) method. We set the integration step to minutes to match the simulation time step. The Neural ODE thereby learns the building's thermodynamic

behavior in continuous time and, for multi-step prediction, repeatedly applies the RK4 method. In our implementation, the model performs a 96-step (24-hour) rollout using the Neuromancer library (Dr̄goña et al., 2023).

Differentiable Predictive Control (DPC)

The primary contribution of this work is the application of DPC to whole-building multi-zone HVAC control. DPC models unknown nonlinear dynamics and learns a constrained optimal control policy. DPC is an unique MPC alternative to conventional MPC (Dr̄goña et al., 2022). Unlike conventional MPC, which solves a real-time optimization at every time step, DPC leverages deep learning–based automatic differentiation to develop a control policy based on previous simulated or operational data.

In this work, we train a simple building dynamics model based on a Neural ODE and embed it as the predictor in a differentiable closed loop. The model relies solely on a limited set of measurable parameters, including zone and outdoor air temperature. A neural network parameterizes the control policy π_θ ; its inputs and outputs are defined in Equation (3).

$$u_k = \pi_\theta(x_k, d_k, T_{in, min,k}, T_{in, max,k}) \quad (3)$$

Here, $T_{min,k}$ and $T_{max,k}$ denote the vector-valued thermal comfort bounds at time k . Also, we use Gaussian Error Linear Units (GELUs) as an internal activation to capture nonlinearities. The policy network is an MLP with two hidden layers, each with 64 neurons. During training, we used up to 300 epochs with early stopping and a patience of 10 based on validation performance, and we applied a warm-up period of 50 epochs to improve training stability. This architecture enables the policy to simultaneously learn the nonlinear system dynamics and operational constraints directly from simulated time-series data across diverse operating conditions—without a supervisory controller.

We formulate the problem over a prediction horizon N as minimizing a cost function that includes (i) a state tracking term, (ii) a smoothing term that penalizes the rate of change of the control input to discourage excessive switching and HVAC cycling, and (iii) total cooling-energy use. The objective and constraints are given in equation (4).

$$\min_{u_k} \sum_{k=1}^N (\alpha \|x_k - x_{ref}\|_2^2 + \beta \|u_k - u_{k-1}\|_2^2 + \gamma \|E_k\|_2^2) \quad (4)$$

Subject to (2), (3)

Here, x_{ref} is reference temperature, E_k represents the total HVAC system energy consumption, including cooling, supply fan, and reheat. For the objective in Eq. (4), we set the weighting coefficients for tracking loss = 1000, control loss = 0.1, and energy loss = 0.01. The optimization uses time-series data sampled from a distribution spanning diverse initial conditions, outdoor conditions, and occupancy scenarios. The optimization model is trained via stochastic gradient descent with the AdamW optimizer. The resulting DPC policy satisfies indoor thermal-comfort constraints while achieving energy-efficient control performance.

Case Study

Building and HVAC System Overview

The training and validation datasets for our DPC implementation were generated with EnergyPlus. The testbed is a high-fidelity model of the DOE Prototype Medium Office building developed in prior work.

Figures 3 and 4 show the building section and zoning layout. This medium-sized office is a three-story, rectangular building (161 ft $\times$ 108 ft) with 17,420 ft² per floor (52,260 ft² total). It comprises 12 space types—including open offices, private offices, conference rooms, and lobby—and is aggregated into five multi-zones for control and evaluation. Multiple spaces are grouped into the same thermal zone and governed by a single thermostat; this zoning is mirrored in the BEM.

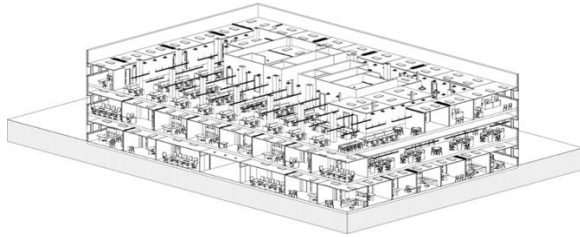


Figure 3. High-Fidelity Medium-Sized Office Building

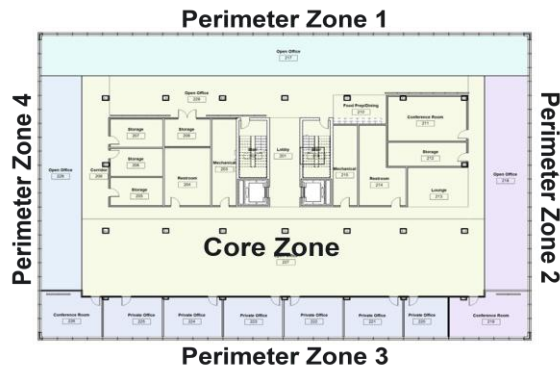


Figure 4. Building Zoning Overview

Table 2 summarizes the key characteristics of the target building. Internal gains (occupancy, plug loads, and lighting) are specified in accordance with ASHRAE 62.1 and the Commercial Buildings Energy Consumption Survey (CBECS). The lighting design includes more than 800 luminaires—combining troffers, linear pendants, and downlights—to meet task illuminance. Illuminance targets follow the IES Lighting Handbook (10th ed.), and lighting power density and control requirements comply with ASHRAE Standard 90.1–2019.

Table 2. Target Building Overview

Categories	Description
Building Information	Total area: 52,260 ft ² (Each floor: 17,420 ft ²)
	Slab on grade (F-Factor): 0.73
Construction Performance	Roof (U-Factor): 0.063
	Wall (U-Factor): 0.084
	Window (U-Factor): 0.57
	Window (SHGC): 0.39
	Door (U-Factor): 0.7
Internal Loads	Plug Loads (W/ft ²): 0.598
	Occupancy (People/ft ²): 0.0075
	Lighting (W/ft ²): 1.03

The HVAC system (see Figure 5) is a packaged VAV with reheat rooftop unit compliant with ASHRAE 90.1–2019 PRM (Appendix G). One RTU per floor supplies five zones via ductwork, including a plenum, for a total of 15 zones. Each RTU includes a variable-speed supply fan and DX cooling/heating coils; each VAV terminal comprises a reheat coil and a modulating damper. All zone cooling setpoints are 24 °C with a setback temperature of 26.7 °C.

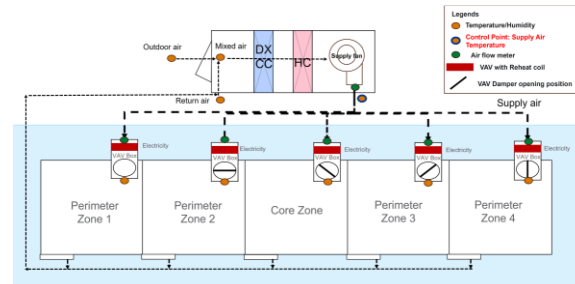


Figure 5. Multi-Zone VAV Systems

Stochastic Occupancy Scenario

We generate occupancy schedules using an agent-based probabilistic occupancy simulator proposed by Chen et al. (2016). The tool combines a survey-based schedule

library with a state-transition model to produce stochastic estimates of occupancy status and occupant count at 15-minute intervals. States are modeled as absent $\leftrightarrow$ present $\leftrightarrow$ away, with event timing and durations (e.g., arrivals, departures, lunch, meetings) drawn from distributions (normal, log-normal, discrete mixtures). Heterogeneity by space type (open office/private office/meeting room), floor (bottom, middle, top), and location (core/perimeters) is parameterized.

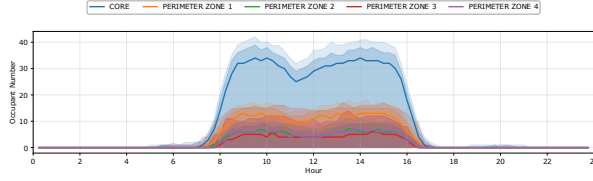


Figure 6. Probabilistic occupancy profiles for the bottom floor

Figure 6 illustrates the stochastic occupancy distribution for the five HVAC zones on the bottom floor as a representative example. The bold line denotes the median occupant count per zone; the dark band shows the interquartile range (25–75%), and the light band indicates the 10–90% quantiles. CORE_BOT exhibits the largest mean and peak with the widest variability, whereas PERIMETER_BOT zones show lower medians and relatively higher dispersion, reflecting solar and workstation effects. These time-varying patterns make explicit the disturbance characteristics needed for robust DPC evaluation.

Datasets

The test building was simulated in Climate Zone 4C. To intentionally capture seasonal effects, we introduce distributional differences across the datasets. Figure 7 compares the outdoor dry-bulb temperature distributions during the cooling season (May–September) for the training, validation, and test sets.

- The training period (May–July) covers a broad temperature spectrum—from the milder early-summer conditions in May to the relatively higher temperatures in July.
- The validation period (August) contains consistently higher temperatures.
- The test period (September) covers an early-fall transition, with fewer hot hours and more moderate temperatures.

Co-simulation setup

We developed a co-simulation framework that couples the virtual control platform, EnergyPlus, with an external optimal controller DPC via the Python API in real time.

EnergyPlus simulates indoor thermal conditions and HVAC operation at 15-minute intervals, while the Python environment repeatedly solves the DPC optimization using a pre-trained Neural ODE predictor. Figure 8 illustrates the data flow for the Neural-ODE-based DPC.

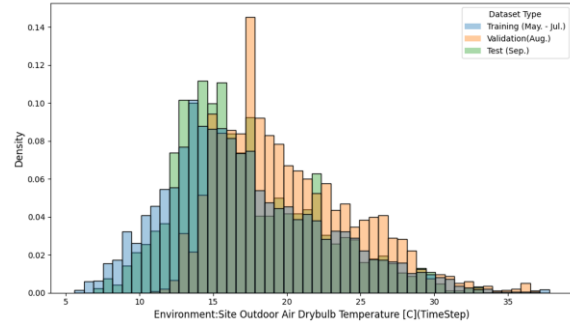


Figure 7. Distribution of outdoor air dry-bulb temperature by dataset split

In the Input data stage, building and weather data are gathered. At the start of each time step, EnergyPlus calls the runtime API callback `callback_begin_system_timestep_before_predictor`, which invokes a state variable with the current simulation state. Within this function, real-time state variables are read and passed to the DPC for multi-step prediction and optimization.

In the Predictor stage, the Neural ODE model forecasts future states. The constraints specify indoor thermal comfort requirements and equipment limits. In the optimization stage, the DPC controller computes the optimal control action. During the EnergyPlus stage, the runtime API callback `callback_after_predictor_before_hvac_managers` is executed at the end of the predictor step but before the HVAC managers. It overwrites the relevant actuators with the optimized control input.

Finally, in the updating output data stage, the resulting simulation outputs are feedback as inputs for the next time step’s prediction and optimization. These callbacks are part of the EnergyPlus runtime API, which allows a Python client to sense and actuate data in a running simulation via sensor queries and actuator overrides on the current state.

This closed-loop, receding-horizon control is maintained over the entire simulation, enabling quantitative evaluation of controller performance under diverse occupancy patterns and load conditions.

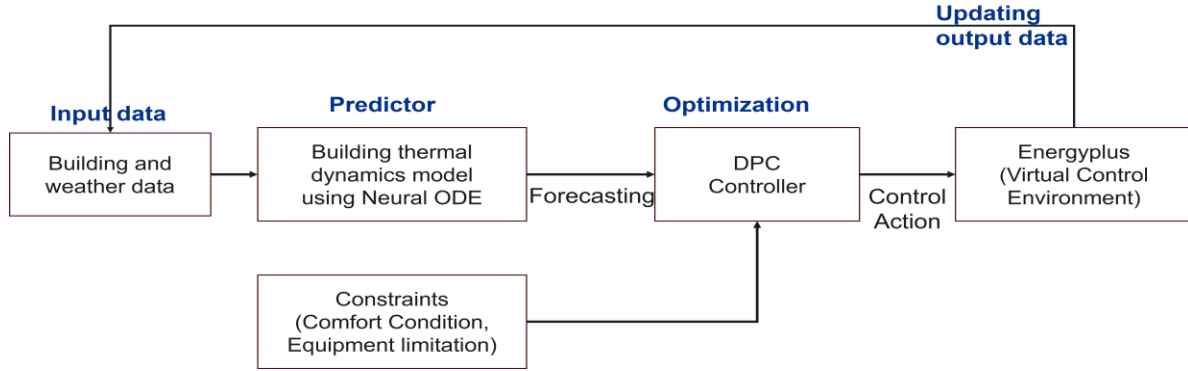


Figure 8. Data Flow of Neural ODE-based DPC

Results and Analysis

Prediction Performance evaluation

To evaluate the predictive performance of the DPC, we assessed single-zone accuracy on the test data set for the bottom-floor core zone—the same zone used during training. Figure 9 shows time-series tracking of measured versus predicted temperature (left) alongside the corresponding parity plots (right); the traces align closely across both transient and steady periods, and the point clouds cluster near the 1:1 line, indicating minimal systematic bias. Quantitative performance is Training (May–July): MAE: 0.092 °C; CVRMSE: 0.69%; R^2 : 0.979; Validation (August): MAE: 0.110 °C; CVRMSE: 0.79%; R^2 : 0.974; Test (September): MAE: 0.108 °C; CVRMSE: 0.78%; R^2 : 0.974. These results indicate a small generalization gap between seen (training) and unseen (validation/test) conditions. In other words, the predictor maintains high accuracy across shifts in outdoor air temperature distributions, demonstrating robust generalization performance.

Building on this result, we evaluated the generality and robustness of the DPC by deploying it throughout the three-story medium office building (one RTU per floor; five terminal zones per floor; 15 zones total). The Neural-ODE predictor was trained once using May–July data from the bottom-floor core zone (CORE_BOT) and then applied unchanged—with no additional training, tuning, or calibration—to the remaining 14 zones. Preprocessing (normalization) was fixed to CORE_BOT statistics and used consistently across all evaluations; hyperparameters were likewise fixed a priori. Performance was assessed separately for August (validation) and September (test) using MAE (°C), CVRMSE (%), and R^2 .

Table 3 summarizes results by floor and zone. Across all core zones, the test-set performance was highly stable—CVRMSE: at most 0.83%; MAE: at most 0.129 °C; R^2 : at least 0.973. Many perimeter zones also maintained

strong accuracy—CVRMSE: 0.88–1.76%; MAE: 0.14–0.24 °C; R^2 : 0.90–0.97—indicating that a model trained on a single zone can primarily preserve predictive fidelity and control suitability under changes in season, floor, and zone. An exception was PERIMETER_BOT_ZN_3 in September, where performance degraded—CVRMSE: 2.92%; MAE: 0.267 °C; R^2 : 0.692. As suggested by Figure 3, this zone comprises mainly conference rooms and private offices; under hybrid work patterns, meetings and event-driven occupancy cluster into short bursts. Consistently, the probabilistic occupancy fan chart in Figure 6 shows wide 10–90% bands around 08:00–10:00 and 14:00–16:00, indicating higher variability and sharper peak loads than in other perimeter zones.

Despite this outlier, the zero-shot transfer to the other 14 untrained zones demonstrates strong overall generalization and robustness. Practically, this reduces the need to build and tune detailed zone-specific models as building scale increases, improving scalability and supporting deployment in commercial buildings with extensive zoning.

SAT Reset Strategies Evaluation

Four SAT reset strategies—Baseline (fixed SAT), OAT Reset, G36 Trim & Respond, and DPC—were compared over the cooling season (May–September).

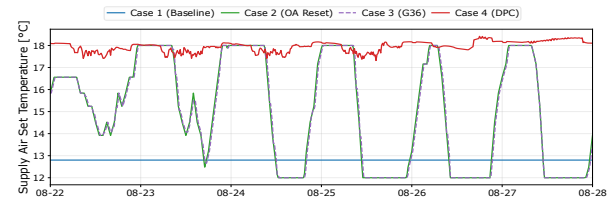


Figure 10. Comparison of SAT Reset Strategies
(Representative Cooling week: Aug. 22–27)

As shown in Figure 10, Baseline fixed SAT at 12.8 °C, OAT Reset swings widely with outdoor conditions (12–18 °C), and G36 often drives SAT to the lower bound

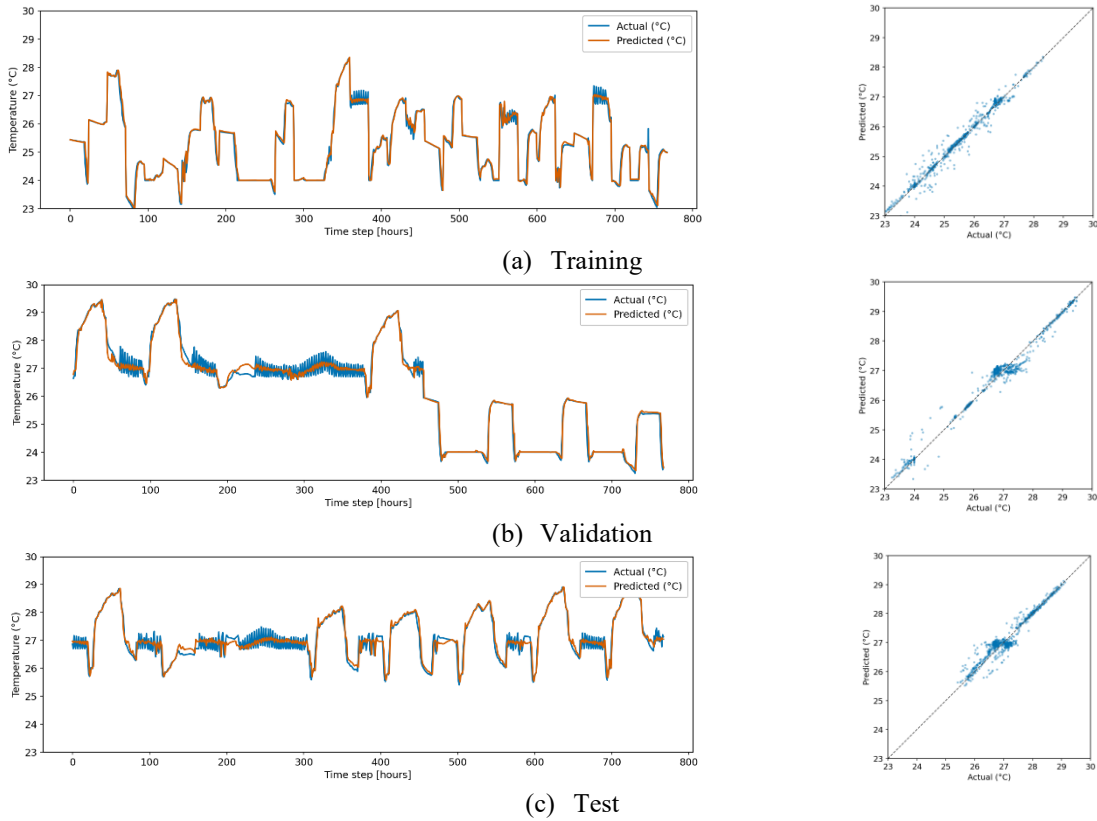


Figure 9. Representative zone temperature prediction (left) and parity plots (right) for equal portions of the Training, Validation, and Test periods.

(12.8 °C) during peaks, risking prolonged saturation and over-cooling. DPC, by contrast, keeps SAT in a narrow band near 18 °C with brief, targeted dips only when needed, yielding the most stable and predictive behavior among the four strategies. Taken together, these SAT-reset results suggest that the G36 logic may struggle to operate effectively under probabilistic, highly imbalanced occupancy conditions. In our simulations, realistic lighting loads and stochastic office-type occupancy patterns create pronounced load discrepancies between core and perimeter zones (Fig. 6). Under these conditions, zone-level demand signals become noisy and uneven, and the T&R mechanism tends to overreact or saturate at the lower SAT bound, leading to overcooling and, in several cases, worse performance than the more straightforward outdoor-air-reset strategy. tends to overreact or saturate at the lower SAT bound, leading to overcooling and, in several cases, worse performance than the more straightforward outdoor-air-reset strategy.

Figure 11 presents time-series indoor temperatures for the core zone and four perimeter zones on the bottom floor, comparing five SAT reset strategies. Each panel

shows the occupied period (gray background), zone air temperature (solid line), setpoint (dashed line), and the ± 1.0 °C comfort band (yellow shading) for the representative cooling week of August 22–28. The Baseline strategy, which employs a fixed setpoint, fails to adapt to variations in outdoor conditions or occupancy and often overcools. Although the OAT Reset strategy reflects outdoor temperature trends, several zones experience afternoon overshoots and noticeable oscillations, indicating reduced control stability. The G36 strategy frequently drives zone temperatures below the lower comfort bound during occupied peaks, resulting in pronounced overcooling. In contrast, the DPC strategy maintains temperatures smoothly near the setpoints across all zones during occupied periods, demonstrating the most stable and adaptive response among the strategies. Overall, DPC achieved the most consistent performance in maintaining comfort bounds and tracking setpoints, whereas Baseline and G36 showed pronounced overcooling tendencies and limited adaptability to stochastic occupancy variations.

Figure 12 compares zone-level cumulative thermal comfort violations (°C·h) during the representative cooling week (August 22–28, occupied hours 05:00–

21:00) across four SAT reset strategies. A violation represents the time-integrated absolute temperature deviation whenever the zone air temperature departs from the comfort band defined as setpoint ± 1.0 °C.

Across all BOT-floor zones, DPC yields the lowest violations and the largest improvements over Baseline. The Baseline-relative reductions are: CORE 79.7%,

Table 3. Zero-shot transfer performance across floors and zones

Floor	Zone Name	Metrics	Training (May- July)	Validation (August)	Test (September)
Bottom	CORE	MAE (°C)	0.092	0.110	0.108
		CVRMSE (%)	0.69	0.79	0.78
		R ²	0.979	0.974	0.974
	PERIMETER_1	MAE (°C)	0.134	0.170	0.153
		CVRMSE (%)	0.83	0.99	0.96
		R ²	0.969	0.954	0.959
	PERIMETER_2	MAE (°C)	0.198	0.237	0.213
		CVRMSE (%)	1.42	1.66	1.49
		R ²	0.937	0.904	0.935
	PERIMETER_3	MAE (°C)	0.227	0.224	0.267
		CVRMSE (%)	2.49	1.83	2.92
		R ²	0.755	0.855	0.692
Middle	CORE	MAE (°C)	0.211	0.235	0.218
		CVRMSE (%)	1.73	1.76	1.73
		R ²	0.920	0.902	0.918
	PERIMETER_1	MAE (°C)	0.094	0.113	0.121
		CVRMSE (%)	0.69	0.78	0.80
		R ²	0.980	0.974	0.973
	PERIMETER_2	MAE (°C)	0.126	0.161	0.142
		CVRMSE (%)	0.78	0.94	0.88
		R ²	0.976	0.962	0.967
	PERIMETER_3	MAE (°C)	0.150	0.178	0.177
		CVRMSE (%)	0.96	1.08	1.12
		R ²	0.950	0.930	0.930
Top	CORE	MAE (°C)	0.153	0.195	0.203
		CVRMSE (%)	1.08	1.42	1.62
		R ²	0.955	0.922	0.922
	PERIMETER_1	MAE (°C)	0.136	0.162	0.163
		CVRMSE (%)	0.97	1.09	1.16
		R ²	0.949	0.931	0.927
	PERIMETER_2	MAE (°C)	0.108	0.127	0.129
		CVRMSE (%)	0.73	0.83	0.83
		R ²	0.979	0.971	0.973
	PERIMETER_3	MAE (°C)	0.178	0.178	0.177
		CVRMSE (%)	1.41	1.09	1.17
		R ²	0.942	0.951	0.954
	PERIMETER_4	MAE (°C)	0.157	0.179	0.172
		CVRMSE (%)	1.05	1.16	1.15
		R ²	0.947	0.924	0.940
BOT	CORE	MAE (°C)	0.149	0.174	0.176
		CVRMSE (%)	1.04	1.17	1.28
		R ²	0.952	0.929	0.932
	PERIMETER_1	MAE (°C)	0.205	0.204	0.211
		CVRMSE (%)	1.80	1.57	1.64
		R ²	0.909	0.908	0.922

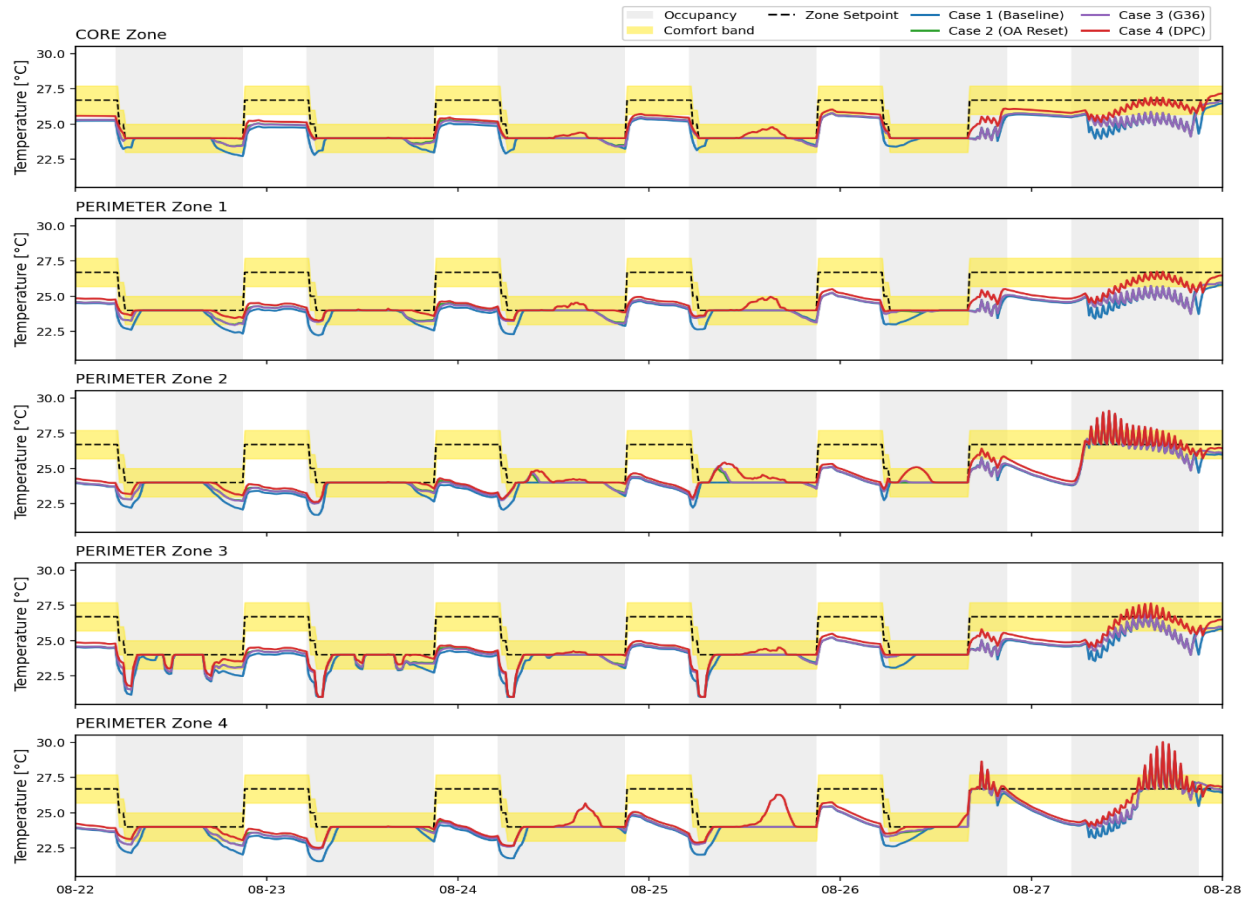


Figure 11. Time-series comparison of zone air temperatures for the core and perimeter zones on the BOT floor under four SAT reset strategies (Representative Cooling week: Aug. 22)

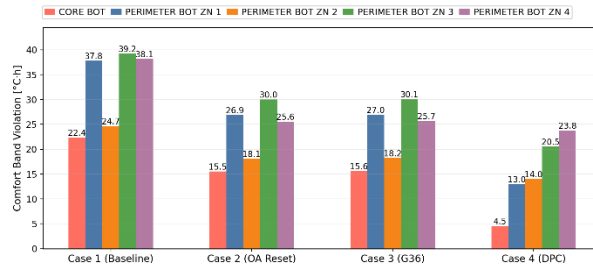


Figure 12. Zone-level cumulative thermal-comfort violations ($^{\circ}\text{C}\cdot\text{h}$) during occupied hours under four SAT reset strategies on the BOT floor (Representative Cooling week: Aug. 22-27)

PER-ZN1 65.7%, PER-ZN2 43.4%, PER-ZN3 47.7%, PER-ZN4 37.6%. By contrast, OA Reset achieves moderate improvements of 23.4–33.0% across zones, and G36 shows similar levels of improvement. Averaged over all five zones, DPC improves violations by 54.8%, indicating superior robustness to stochastic occupancy and load variations and more consistent tracking within the comfort band than conventional reset strategies.

Figure 13 presents a comparison of total energy consumption—comprising cooling, reheat, and fan energy—across four SAT reset control strategies during the representative cooling week. Case 1 (Baseline), which operates with a fixed supply air temperature (SAT), results in the highest overall consumption (2154.2 kWh) due to persistent overcooling and elevated reheat demand.

In contrast, Case 4 (DPC) dynamically adjusts the SAT based on actual indoor load conditions, achieving balanced reductions in all three components. Its total energy consumption is 1695.1 kWh, the lowest among all strategies, corresponding to a 21.3% reduction (459.1 kWh) relative to Baseline. Other rule-based strategies, including Case 3 (G36) and Case 2 (OA Reset), show more modest savings—11.9% (256.1 kWh) and 11.6% (250.7 kWh), respectively. These results highlight the superior performance of prediction-based DPC, which more effectively minimizes total energy use through proactive, load-responsive control, outperforming the conventional reset method.

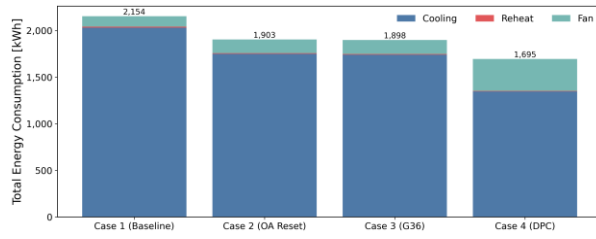


Figure 13. Comparison of Total Energy Consumption (Representative Cooling week: Aug. 22-27)

Figure 14 compares cumulative thermal comfort violations ($^{\circ}\text{C}\cdot\text{h}$) across 15 zones (5 per floor) during the cooling season (May–September) following zero-shot deployment of a Neural ODE-based DPC controller trained only on the Bottom floor. The same controller was applied to the Mid and Top floors without additional tuning.

Total violations across all zones were 7308.1 $^{\circ}\text{C}\cdot\text{h}$ for Baseline, compared to 5099.5 (30.2%) for OA Reset, 5106.3 (30.1%) for G36, and 4436.3 (39.3%) for DPC, the lowest among all strategies. The Bottom-floor CORE zone showed the minimum violation (73.4 $^{\circ}\text{C}\cdot\text{h}$), and significant reductions were observed in both the mid and top floors. These results demonstrate that the DPC controller generalizes effectively across floors, achieving about 39% improvement in comfort without retraining, and highlighting its potential for scalable, whole-building deployment.

Figure 15 presents the total HVAC system energy use (sum of cooling, reheat, and fan energy) for each SAT reset strategy over the entire cooling season (May–September). Compared to Baseline, Case 2 (OA Reset) achieved an 11,154 kWh (8%) reduction, Case 3 (G36) saved 11,442 kWh (8%), and Case 4 (DPC) yielded the most significant savings at 18,695 kWh (14%). These results indicate that DPC is most effective, consistently reducing cooling, reheat, and fan loads in a balanced manner.

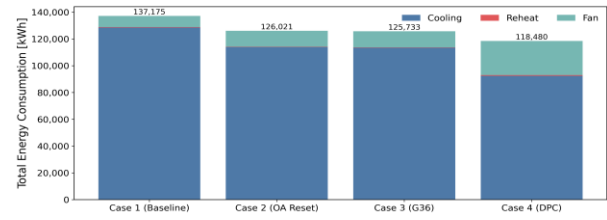


Figure 15. Comparison of Total Energy Consumption (Cooling Season: May. 01–September. 30)

Conclusion

This study demonstrated the effectiveness of a DPC framework for SAT reset in multi-zone commercial buildings with stochastic occupancy. Leveraging a Neural ODE-based predictor, DPC outperformed rule-based strategies—including fixed SAT, OAT reset, and ASHRAE Guideline 36—in both comfort and energy use.

During the representative cooling week, DPC reduced comfort violations by 54.8% and energy use by 21.3% on the bottom floor. Notably, the Neural ODE model was trained only on three months of data from a single zone, rather than on a full cooling season of data across all 15 zones, and even without retraining, the same DPC controller generalized well to upper floors, confirming its zero-shot transferability. Over the full cooling season, DPC achieved the most significant total energy savings—about 14% through balanced reductions in cooling, reheat, and fan energy.

These results highlight DPC as a scalable and occupant-centric control strategy. Future work will explore zone-level extensions to enable supervisory control and further enhance its performance. Other potential areas of investigation include extending the capability to operate without real-time occupancy information and studying the impact of hybrid work schedules on controller performance.

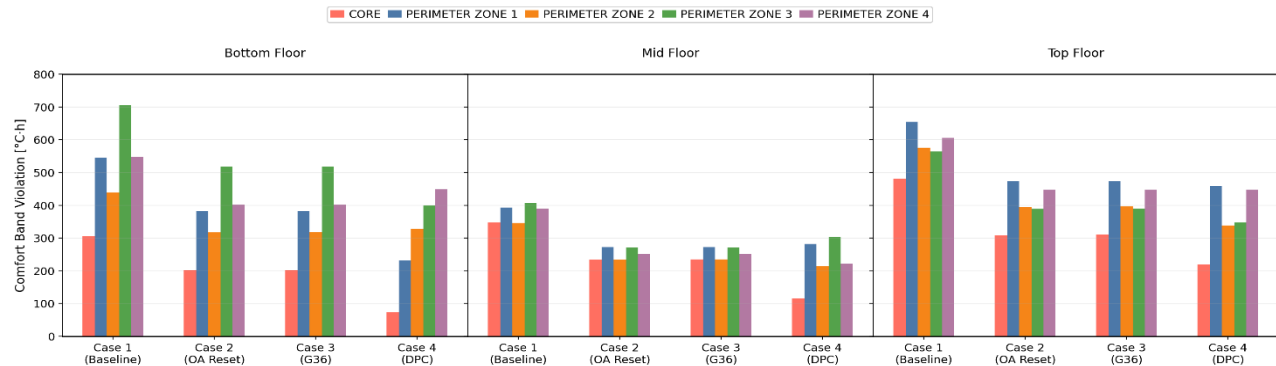


Figure 14. Comparison of Comfort Band Violations by Floor under Supply-Air-Temperature Reset Strategies (Cooling Season: May. 01–September. 30)

AI Disclosure

During the preparation of this work, the authors used ChatGPT (OpenAI, GPT-5.4) for language editing, rephrasing, and improving the clarity of the manuscript. After using this tool, the authors rigorously reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

- O'Brien, W., Wagner, A., Schweiker, M., Mahdavi, A., Day, J., Kjærgaard, M.B., Carlucci, S., Dong, B., Tahmasebi, F., Yan, D., Hong, T., Gunay, H.B., Nagy, Z., Miller, C., Berger, C. 2020. Introducing IEA EBC Annex 79: Key challenges and opportunities in the field of occupant-centric building design and operation, *Building and Environment*, 178, 106738.
- Elehwany, H., Gunay, B., Ouf, M., Cotrufo, N., Venne, J.-S. 2024. Evaluating Common Supply Air Temperature Setpoint Reset Strategies with Varying Occupancy Patterns and Behaviours, *Building and Environment*, 266, 112129.
- Jiang, Z., Wang, X., Li, H., Hong, T., You, F., Drgoňa, J., Vrabie, D., Dong, B. 2025. Physics-informed machine learning for building performance simulation—A review of a nascent field, *Advances in Applied Energy*, 18.
- Drgoňa, J., Kiš, K., Tuor, A., Vrabie, D., Klaučo, M. 2022. Differentiable predictive control: Deep learning alternative to explicit model predictive control for unknown nonlinear systems, *Journal of Process Control*, 116, 80–92.
- Drgoňa, J., Tuor, A., Skomski, E., Vasisht, S., Vrabie, D. 2021. Deep learning explicit differentiable predictive control laws for buildings, *IFAC-PapersOnLine*, 54(6), 14–19.
- Liang, W., Quinte, R., Jia, X., Sun, J.-Q. 2015. MPC control for improving energy efficiency of a building air handler for multi-zone VAVs, *Building and Environment*, 92, 256–268.
- Lee, J.M., Hong, S.H., Seo, B.M., Lee, K.H. 2019. Application of artificial neural networks for optimized AHU discharge air temperature set-point and minimized cooling energy in VAV system, *Applied Thermal Engineering*, 153, 726–738.
- Yoon, Y., Amasyali, K., Li, Y., Im, P., Bae, Y., Liu, Y., Zandi, H. 2024. Energy performance evaluation of the ASHRAE Guideline 36 control and reinforcement learning-based control using field measurements, *Energy & Buildings*, 325, 115005.
- Fang, X., Gong, G., Li, G., Chun, L., Peng, P., Li, W., Shi, X., Chen, X. 2022. Deep reinforcement learning optimal control strategy for temperature setpoint real-time reset in multi-zone building HVAC system, *Applied Thermal Engineering*, 212, 118552.
- Zou, Z., Yu, X., Ergan, S. 2020. Towards optimal control of air handling units using deep reinforcement learning and recurrent neural network, *Building and Environment*, 168, 106535.
- Chen, Y., Luo, X., Hong, T. 2016. An Agent-Based Occupancy Simulator for Building Performance Simulation, LBNL-1004492, Lawrence Berkeley National Laboratory (LBNL).
- Yang, T., Sangogboye, F. C., Arendt, K., Filonenko, K., Dallaire, J., Kjærgaard, M. B., & Veje, C. (2021). The Impact of Occupancy Prediction Accuracy on the Performance of Model Predictive Control (MPC) in Buildings. *Proceedings of the 17th IBPSA Conference*
- Mantesi, E., Chmutina, K., Goodier, C. 2022. The office of the future: Operational energy consumption in the post-pandemic era, *Energy Research & Social Science*, 87, 102472.
- McCord, K., Bixler, T.S., Johnston, K., Gupta, T., Poplawski, M., Hickcox, K. 2025. Lighting systems integration with whole-building life cycle assessment, *Energy & Buildings*, 335.
- Xu, W., Dehwah, A.H.A., Collier, J., Poplawski, M., Zhang, J. 2024. Evaluating the Feasibility of a Workflow for Following the ASHRAE Standard 90.1 Performance Rating Method Using Building Information Modeling, *Proceedings of SimBuild 2024: Eleventh National Conference of IBPSA-USA*, Denver, CO, USA, May 21–23, 2024.
- Yan, D., Hong, T., Dong, B., Mahdavi, A., D'Oca, S., Gaetani, I., Feng, X. 2017. IEA EBC Annex 66: Definition and simulation of occupant behavior in buildings, *Energy and Buildings*, 156, 258–270.
- ASHRAE. 2018. ASHRAE Guideline 36-2018: High-Performance Sequences of Operation for HVAC Systems, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA, USA.
- Chen, R.T.Q., Rubanova, Y., Bettencourt, J., Duvenaud, D. 2018. Neural Ordinary Differential Equations, *Proceedings of the 32nd Conference on Neural Information Processing Systems (NeurIPS 2018)*, Montréal, Canada.
- Drgoňa, J., Tuor, A., Koch, J., Shapiro, M., Jacob, B., Vrabie, D. 2023. NeuroMANCER: Neural Modules with Adaptive Nonlinear Constraints and Efficient Regularizations, GitHub repository, Pacific Northwest National Laboratory (PNNL)